In [91]:
```
import pandas as pd #data manipulation library
import seaborn#visualization library for statistical visualization
import seaborn as sns
import matplotlib.pyplot as plt #visualization library
from sklearn.cluster import KMeans
import warnings
warnings.filterwarnings('ignore')
```

In [92]:
```
df = pd.read_csv("C:/Users/rohit/Downloads/Mall_Customers.csv")
```

In [93]:
```
df.head() #preview of the dataset
```

Out[93]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Univariate Analysis

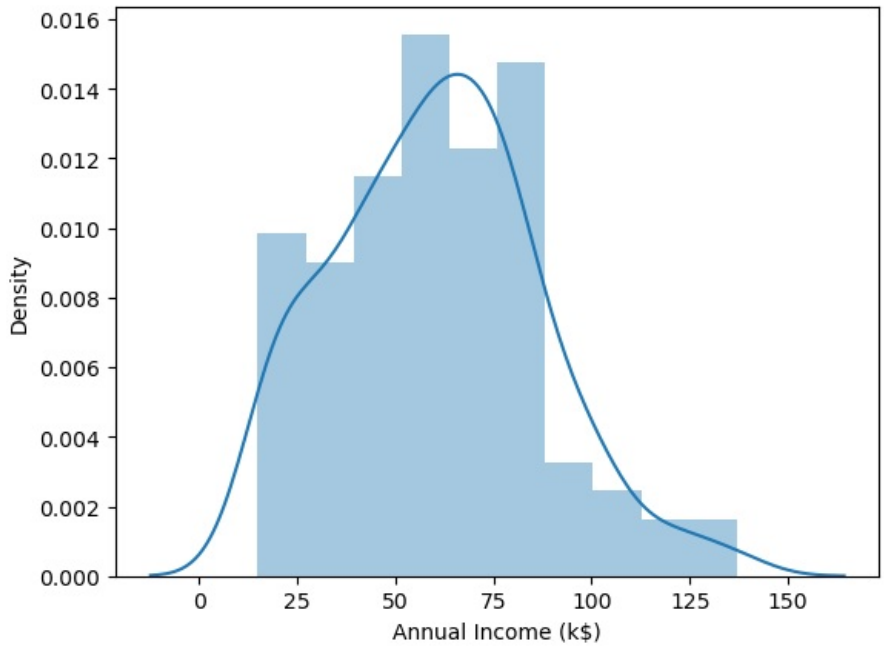
In [94]:
```
df.describe() #the count of the data
```

Out[94]:

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

In [95]:
```
sns.distplot(df['Annual Income (k$)'])
```

Out[95]:
```
<Axes: xlabel='Annual Income (k$)', ylabel='Density'>
```

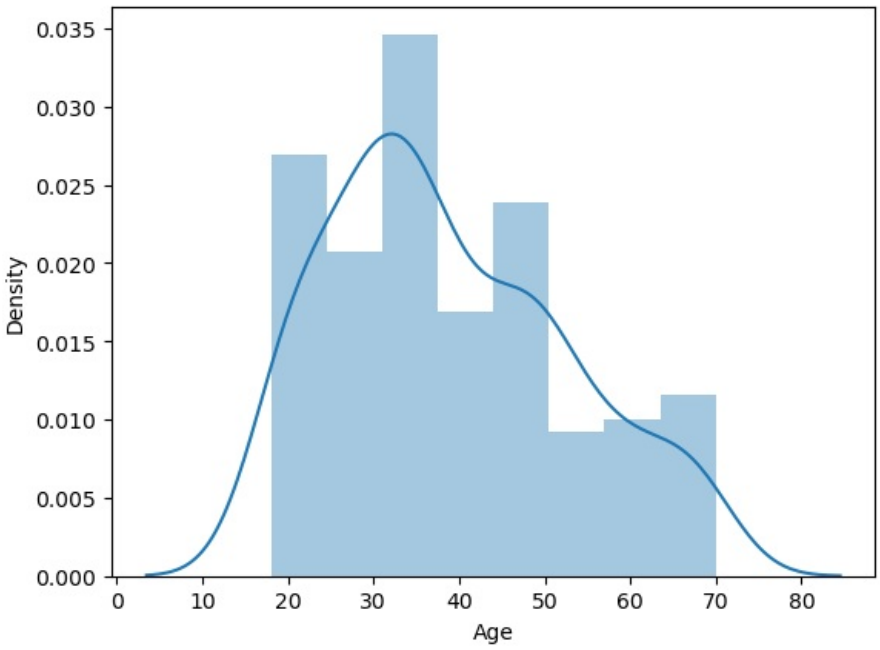


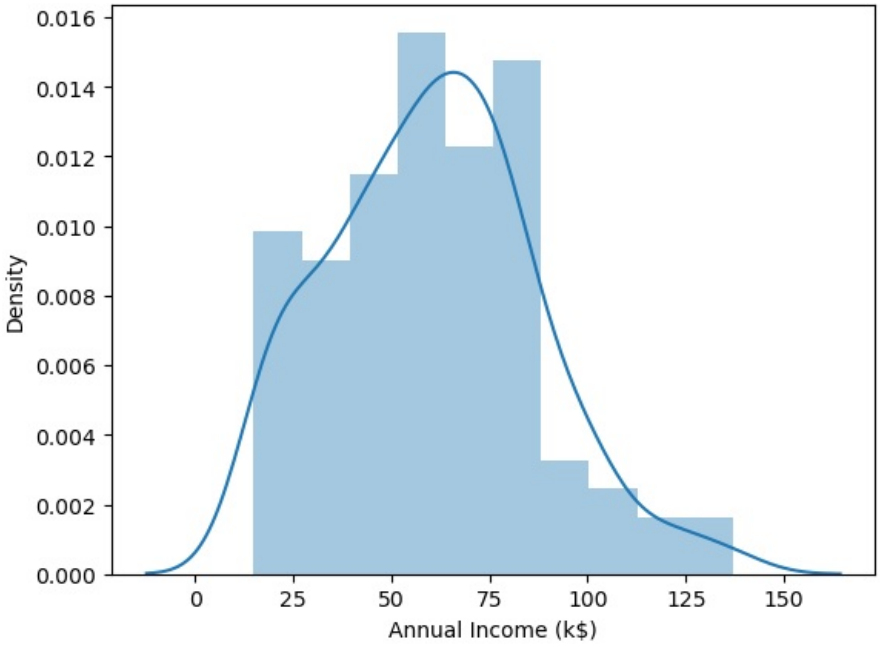
In [96]:
```
df.columns
```

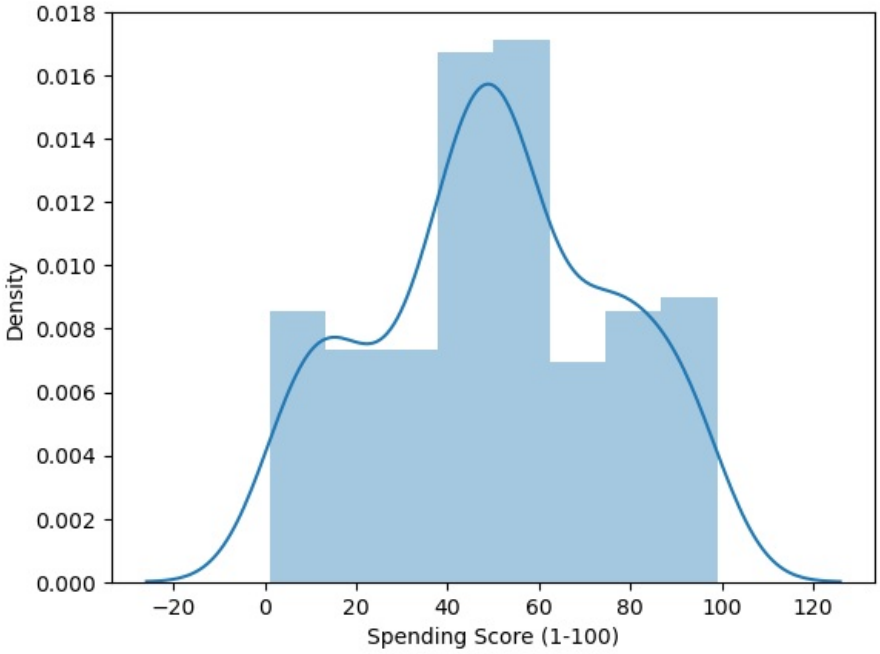
Out[96]:
```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
       'Spending Score (1-100)'],
      dtype='object')
```

In [97]:
```
columns = ['Age', 'Annual Income (k$)',
```

```
          'Spending Score (1-100)']
for i in columns:
    plt.figure()
    sns.distplot(df[i])
```

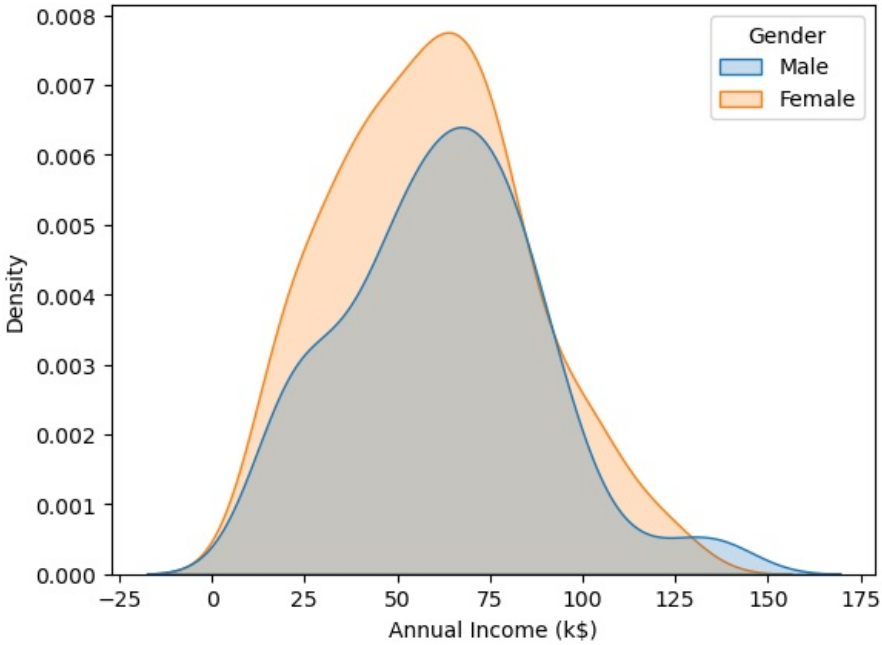




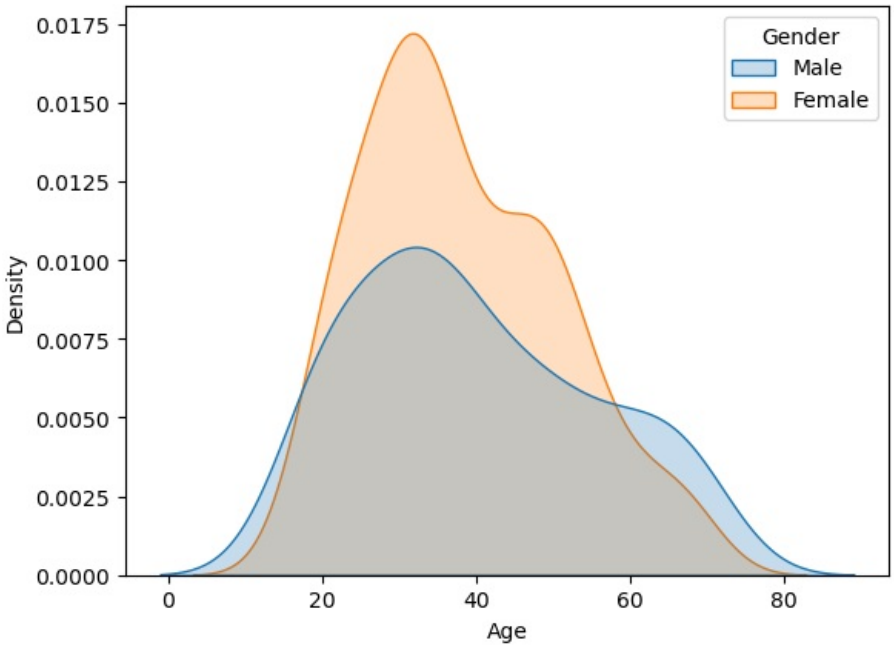


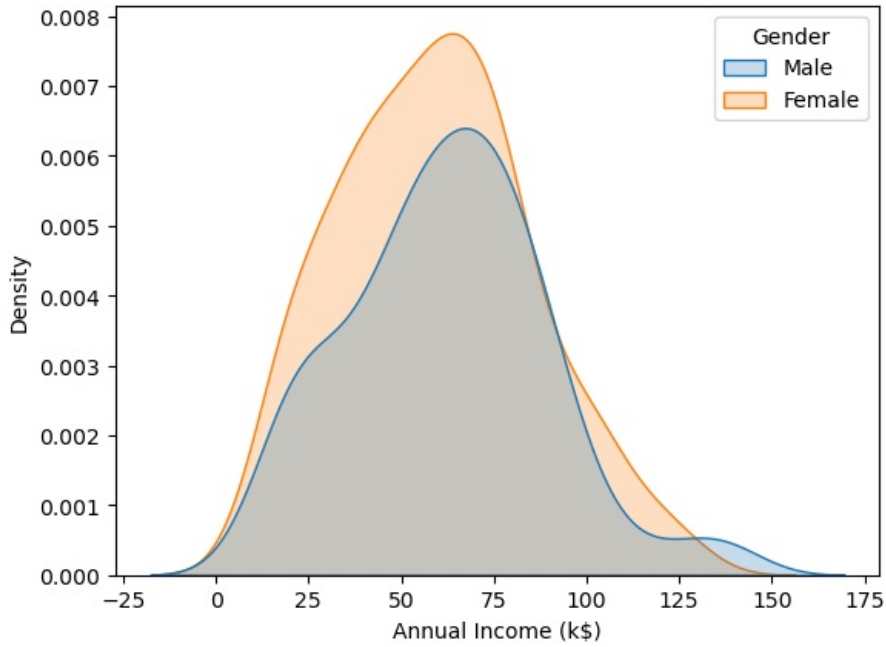
In [98]:
```
sns.kdeplot(data=df, x='Annual Income (k$)', shade=True, hue='Gender')
```

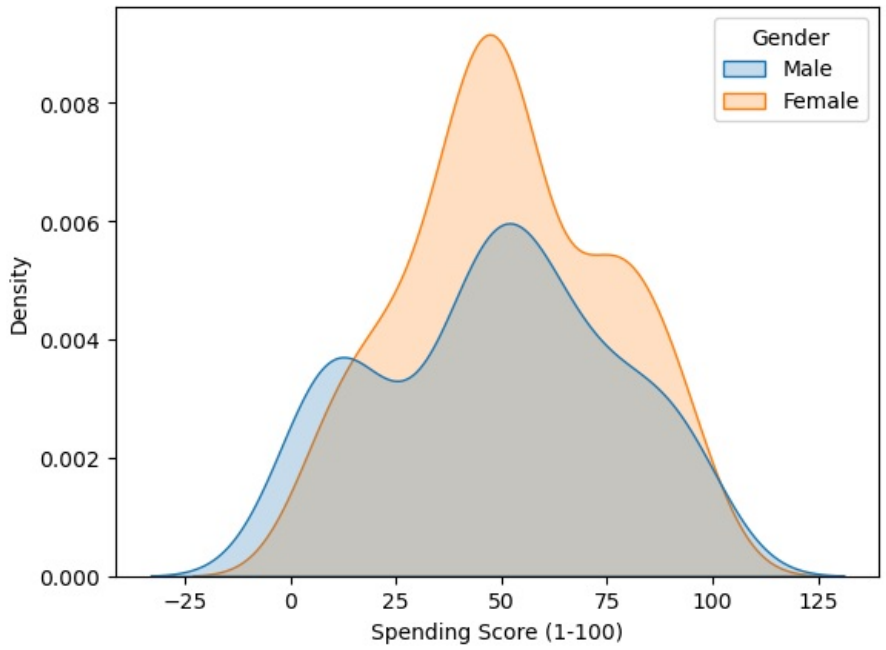
```
plt.show()
```



In [99]:
```
columns = ['Age', 'Annual Income (k$)', 'Spending Score (1-100)']
for i in columns:
    plt.figure()
    sns.kdeplot(data=df, x=i, shade=True, hue='Gender')
```

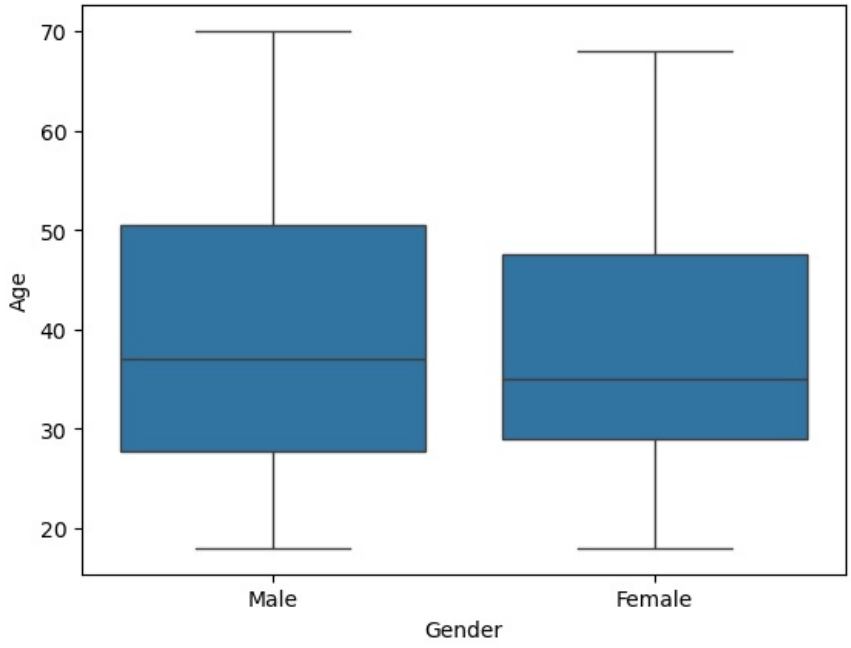


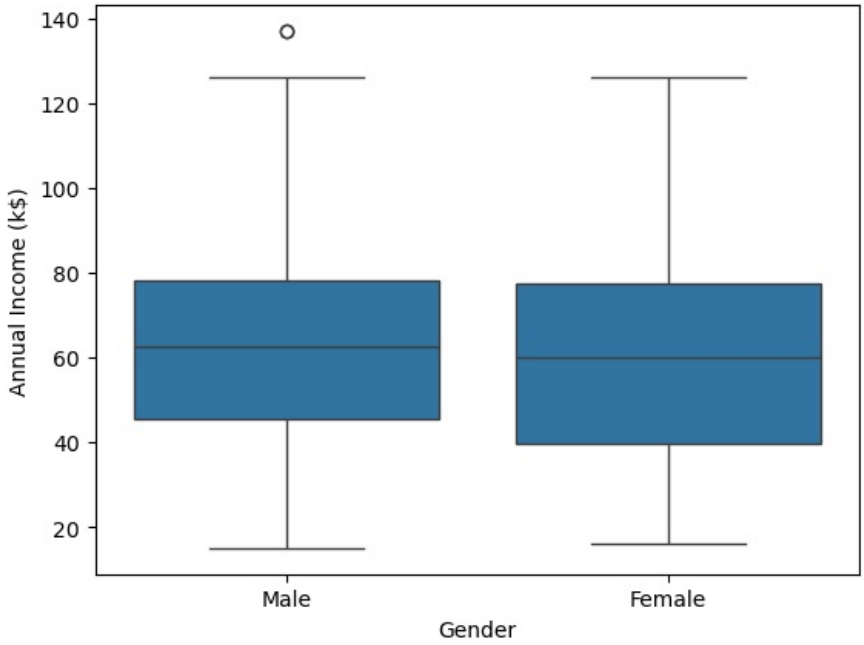


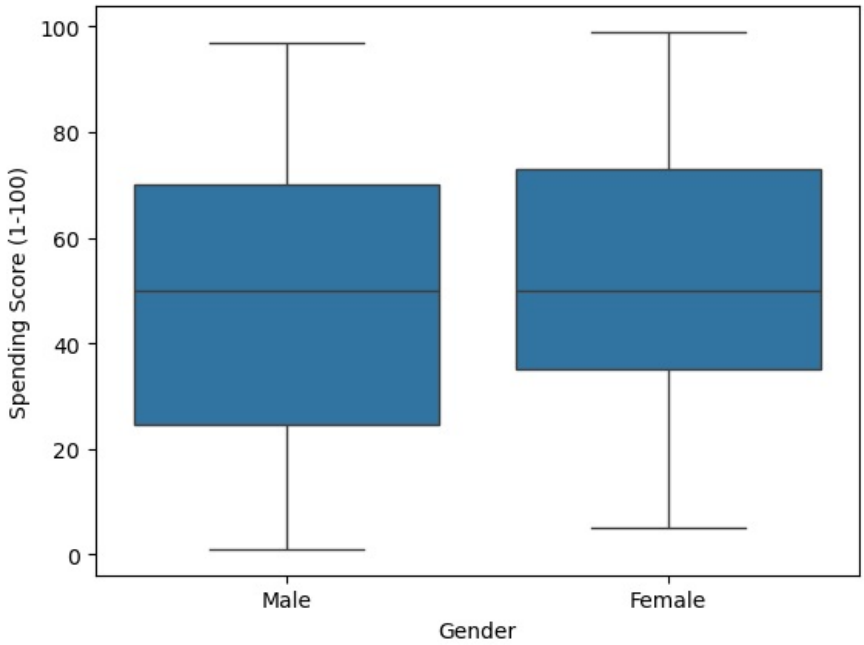


In [100…

```
columns = ['Age', 'Annual Income (k$)', 'Spending Score (1-100)']
for i in columns:
    plt.figure()
    sns.boxplot(data=df, x='Gender',y=df[i])
```







In [101…

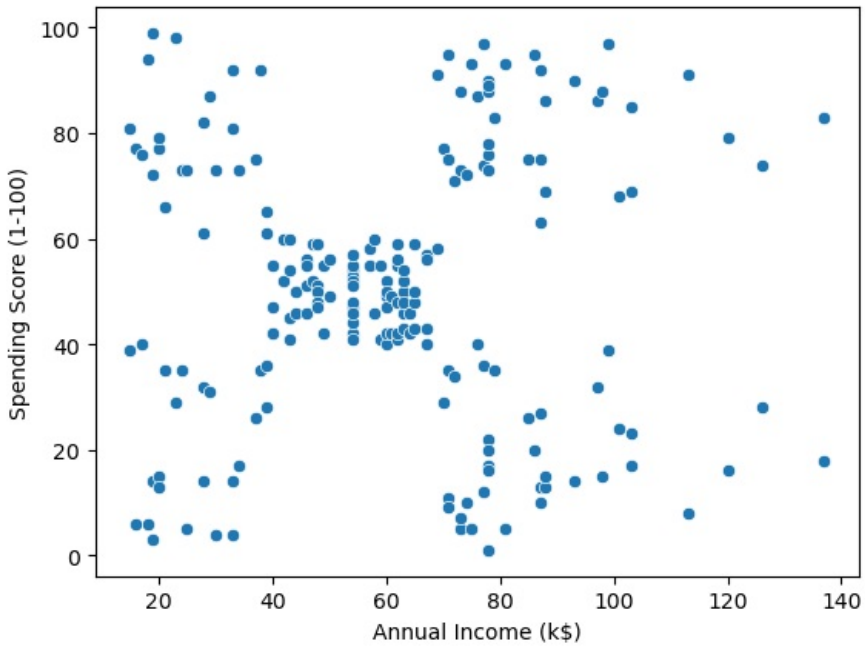
```
df['Gender'].value_counts(normalize=True)
```

Out[101…

```
Gender
Female    0.56
Male      0.44
Name: proportion, dtype: float64
```

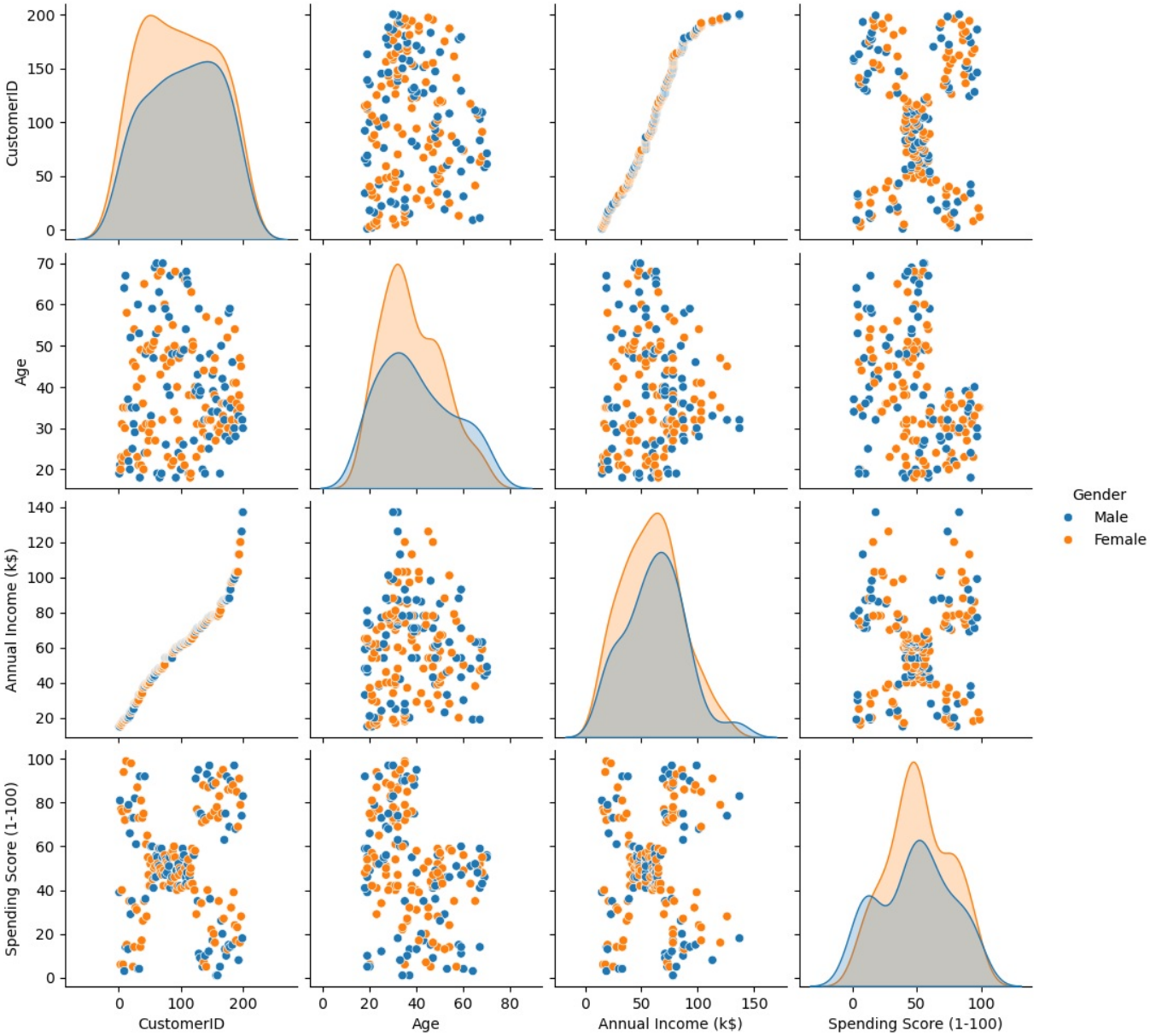
In [102…
```
sns.scatterplot(data=df, x='Annual Income (k$)' ,y='Spending Score (1-100)' )
```

Out[102… <Axes: xlabel='Annual Income (k$)', ylabel='Spending Score (1-100)'>



In [103…
```
#df = df.drop('CustomerID', axis=1)
sns.pairplot(df,hue='Gender')
```

Out[103… <seaborn.axisgrid.PairGrid at 0x166310bd6d0>



In [104…

```
df.groupby(['Gender'])[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']].mean()
```

Out[104…

	Age	Annual Income (k$)	Spending Score (1-100)
Gender			
Female	38.098214	59.250000	51.526786
Male	39.806818	62.227273	48.511364

In [105…

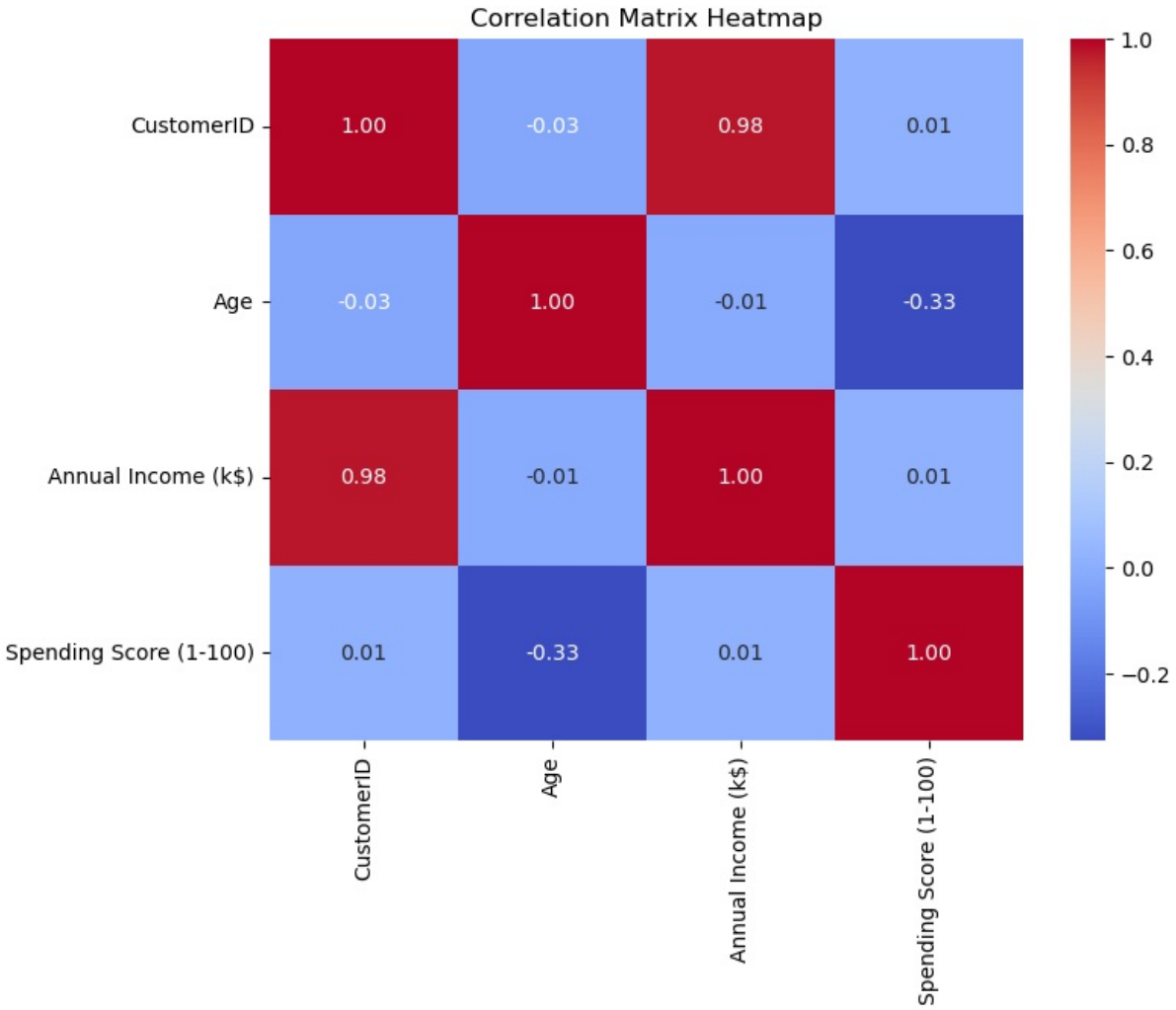
```
print("Data types:")
print(df.dtypes)

correlation_matrix = df.select_dtypes(include=['number']).corr()
print("\nCorrelation Matrix:")
print(correlation_matrix)
```

```
Data types:
CustomerID                 int64
Gender                    object
Age                        int64
Annual Income (k$)         int64
Spending Score (1-100)     int64
dtype: object

Correlation Matrix:
                        CustomerID       Age  Annual Income (k$)  \
CustomerID                1.000000 -0.026763            0.977548
Age                      -0.026763  1.000000           -0.012398
Annual Income (k$)        0.977548 -0.012398            1.000000
Spending Score (1-100)    0.013835 -0.327227            0.009903

                        Spending Score (1-100)
CustomerID                            0.013835
Age                                  -0.327227
Annual Income (k$)                    0.009903
Spending Score (1-100)                1.000000
```

In [106…

```
print("Data types:")
print(df.dtypes)

correlation_matrix = df.select_dtypes(include=['number']).corr()
print("\nCorrelation Matrix:")
print(correlation_matrix)

plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix Heatmap')
plt.show()
```

```
Data types:
CustomerID                 int64
Gender                    object
Age                        int64
Annual Income (k$)         int64
Spending Score (1-100)     int64
dtype: object

Correlation Matrix:
                        CustomerID       Age  Annual Income (k$)  \
CustomerID                1.000000 -0.026763            0.977548
Age                      -0.026763  1.000000           -0.012398
Annual Income (k$)        0.977548 -0.012398            1.000000
Spending Score (1-100)    0.013835 -0.327227            0.009903

                        Spending Score (1-100)
CustomerID                            0.013835
Age                                  -0.327227
Annual Income (k$)                    0.009903
Spending Score (1-100)                1.000000
```



Clustering - Univariate, Bivariate, Multivariate

In [107…
```
from sklearn.cluster import KMeans

clustering1 = KMeans(n_clusters=3)
```

In [108…
```
clustering1.fit(df[['Annual Income (k$)']])
```

Out[108…
```
KMeans
KMeans(n_clusters=3)
```

In [109…
```
clustering1.labels_
```

Out[109…
```
array([2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
       2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
       2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1], dtype=int32)
```

In [110…
```
df['Income Cluster'] = clustering1.labels_
```

```
df.head()
```

Out[110…

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Income Cluster
0	1	Male	19	15	39	2
1	2	Male	21	15	81	2
2	3	Female	20	16	6	2
3	4	Female	23	16	77	2
4	5	Female	31	17	40	2

In [111…

```
df['Income Cluster'].value_counts()
```

Out[111…

```
Income Cluster
0    86
2    58
1    56
Name: count, dtype: int64
```

In [112…

```
clustering1.inertia_
```

Out[112…

```
25640.457784396836
```

In [113…

```
inertia_scores = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(df[['Annual Income (k$)']])
    inertia_scores.append(kmeans.inertia_)
```

In [114…

```
intertia_scores=[]
for i in range(1,11):
    kmeans=KMeans(n_clusters=i)
    kmeans.fit(df[['Annual Income (k$)']])
    intertia_scores.append(kmeans.inertia_)
```

In [115…

```
intertia_scores
```

Out[115…

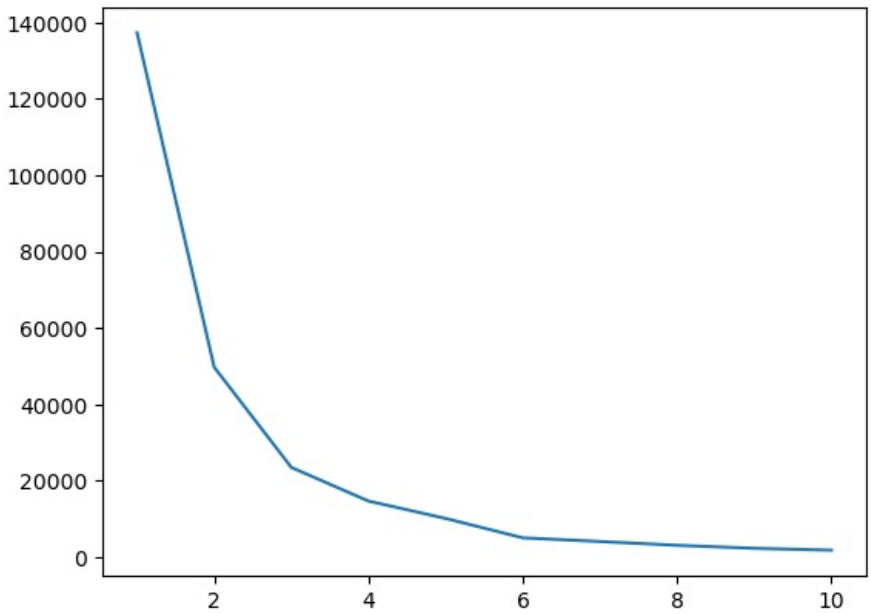
```
[137277.28000000003,
 49761.73701298701,
 23528.152173913048,
 14721.866979949875,
 10175.91862745098,
 5103.8237259816215,
 4143.06481018981,
 3145.603535353535,
 2365.8038073038074,
 1867.8520757020756]
```

In [116…

```
plt.plot(range(1,11),intertia_scores) #elbow starts at 3, therefore we can use 3 as the number of clusters that
```

Out[116…

```
[<matplotlib.lines.Line2D at 0x166357cf390>]
```



In [117…

```
df.columns
```

Out[117…
```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
       'Spending Score (1-100)', 'Income Cluster'],
      dtype='object')
```

In [118…
```
df.groupby('Income Cluster')[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']].mean()
```

Out[118…

	Age	Annual Income (k$)	Spending Score (1-100)
Income Cluster			
0	41.279070	60.906977	50.337209
1	36.910714	92.142857	50.517857
2	37.120690	29.551724	49.689655

Bivariate Clustering

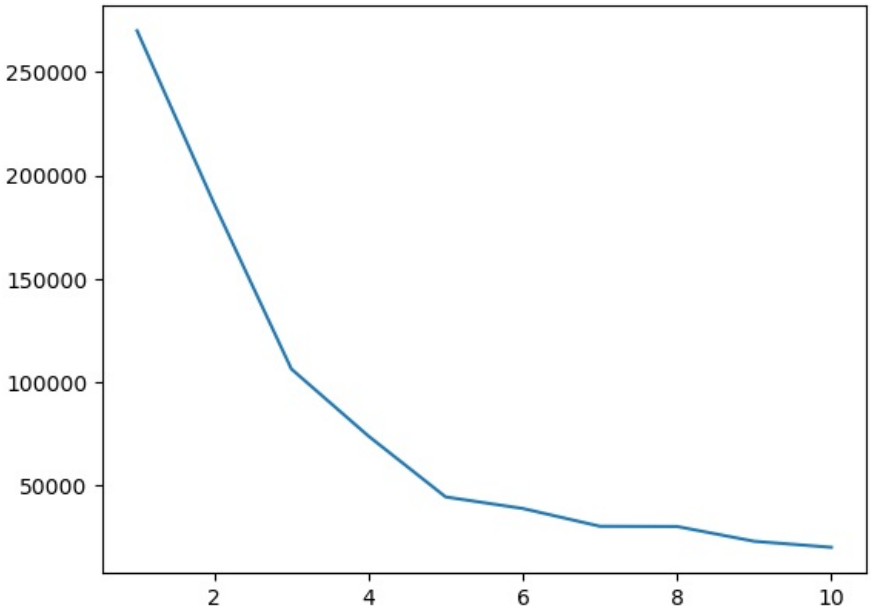
In [119…
```
clustering2 = KMeans()
clustering2.fit(df[['Annual Income (k$)', 'Spending Score (1-100)']])
clustering2.labels_
df['Spending and Income Cluster'] =clustering2.labels_
df.head()
```

Out[119…

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster
0	1	Male	19	15	39	2	4
1	2	Male	21	15	81	2	1
2	3	Female	20	16	6	2	4
3	4	Female	23	16	77	2	1
4	5	Female	31	17	40	2	4

In [120…
```
intertia_scores2=[]
for i in range(1,11):
    kmeans2=KMeans(n_clusters=i)
    kmeans2=KMeans(n_clusters=i)
    kmeans2.fit(df[['Annual Income (k$)', 'Spending Score (1-100)']])
    intertia_scores2.append(kmeans2.inertia_)
plt.plot(range(1,11),intertia_scores2)
```

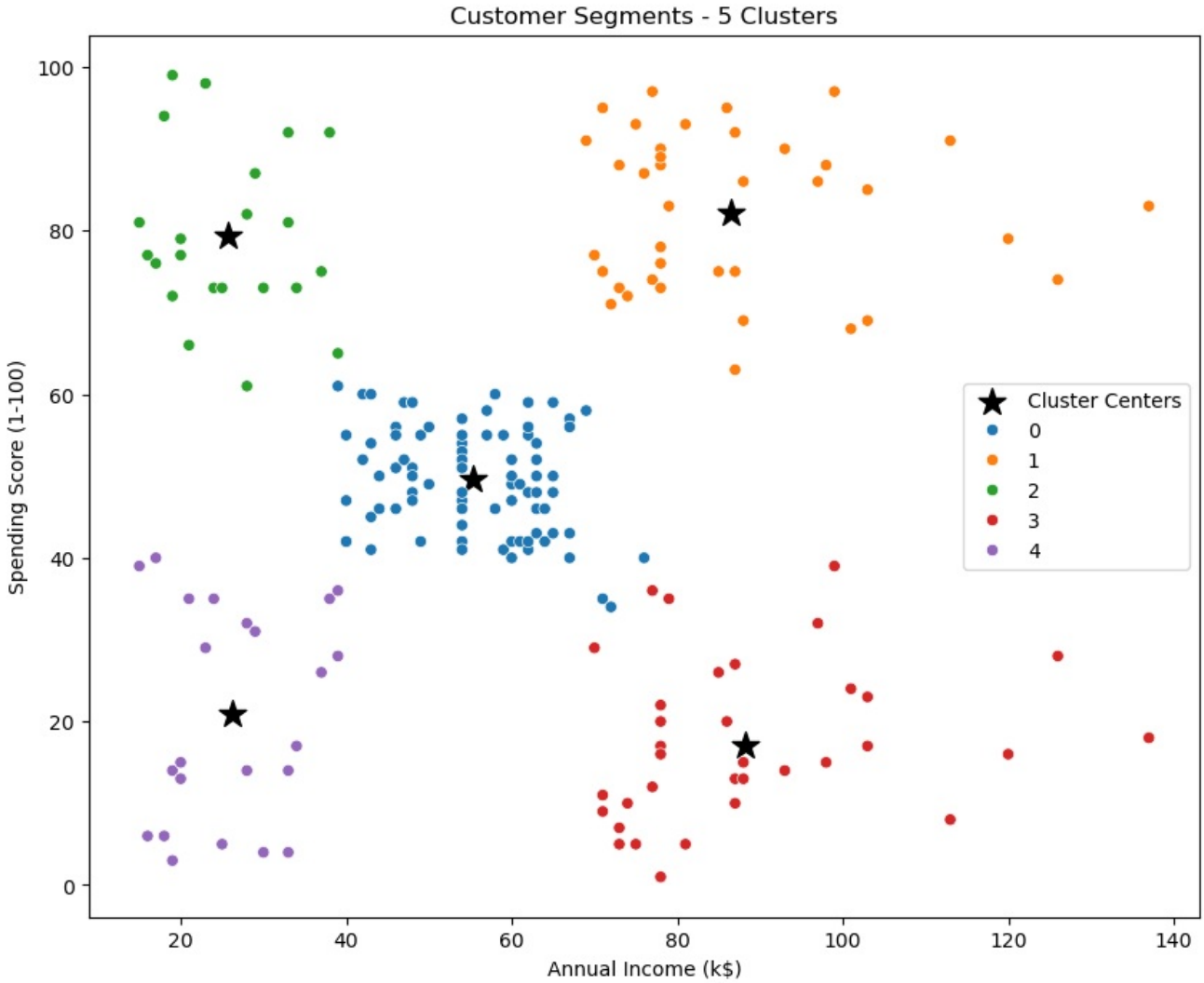
Out[120…
```
[<matplotlib.lines.Line2D at 0x16635606210>]
```



In [121…
```
centers = pd.DataFrame(clustering2.cluster_centers_) #we want to add this x and y to our map, we can do this by
centers.columns = ['x','y']
```

In [122…
```
from sklearn.cluster import KMeans

# Create KMeans with 5 clusters
kmeans_5 = KMeans(n_clusters=5, random_state=42)
df['Spending and Income Cluster'] = kmeans_5.fit_predict(df[['Annual Income (k$)', 'Spending Score (1-100)']])

# Get cluster centers for 5 clusters
centers_5 = kmeans_5.cluster_centers_
centers_df = pd.DataFrame(centers_5, columns=['x', 'y'])
```

```python
# Create the plot with 5 clusters
plt.figure(figsize=(10, 8))

# Plot cluster centers
plt.scatter(x=centers_df['x'], y=centers_df['y'], s=200, c='black', marker='*', label='Cluster Centers')

# Plot data points with 5 clusters
sns.scatterplot(
    data=df,
    x='Annual Income (k$)',
    y='Spending Score (1-100)',
    hue='Spending and Income Cluster',
    palette='tab10'
)

plt.title('Customer Segments - 5 Clusters')
plt.legend()
plt.show()

# Show the cluster centers
print("Cluster Centers for 5 Clusters:")
print(centers_df)
plt.savefig('clustering_bivariate.png')
```




```
Cluster Centers for 5 Clusters:
           x          y
0  55.296296  49.518519
1  86.538462  82.128205
2  25.727273  79.363636
3  88.200000  17.114286
4  26.304348  20.913043
<Figure size 640x480 with 0 Axes>
```

In [123…

```python
crosstab_result = pd.crosstab(
    df['Spending and Income Cluster'],
    df['Gender'],
    normalize='index'
)

print(crosstab_result)
```

```
#percentages
crosstab_result = pd.crosstab(
    df['Spending and Income Cluster'],
    df['Gender'],
    normalize='index'
).round(4) * 100

print("\nGender Distribution by Cluster (%):")
print(crosstab_result)
```

```
Gender                          Female      Male
Spending and Income Cluster
0                             0.592593  0.407407
1                             0.538462  0.461538
2                             0.590909  0.409091
3                             0.457143  0.542857
4                             0.608696  0.391304

Gender Distribution by Cluster (%):
Gender                        Female   Male
Spending and Income Cluster
0                              59.26  40.74
1                              53.85  46.15
2                              59.09  40.91
3                              45.71  54.29
4                              60.87  39.13
```

In [124…

```
df.groupby('Spending and Income Cluster')[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']].mean()
#cluster 1: Ideal for us to run campaigns due to high spending score, high annual income and young age
```

Out[124…

	Age	Annual Income (k$)	Spending Score (1-100)
Spending and Income Cluster			
0	42.716049	55.296296	49.518519
1	32.692308	86.538462	82.128205
2	25.272727	25.727273	79.363636
3	41.114286	88.200000	17.114286
4	45.217391	26.304348	20.913043

In [125…

```
#multivariate clustering
from sklearn.preprocessing import StandardScaler
```

In [126…

```
scale = StandardScaler()
```

In [127…

```
df.head()
```

Out[127…

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster
0	1	Male	19	15	39	2	4
1	2	Male	21	15	81	2	2
2	3	Female	20	16	6	2	4
3	4	Female	23	16	77	2	2
4	5	Female	31	17	40	2	4

In [128…

```
dff = pd.get_dummies(df,drop_first=True)
dff.head()
```

Out[128…

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster	Gender_Male
0	1	19	15	39	2	4	True
1	2	21	15	81	2	2	True
2	3	20	16	6	2	4	False
3	4	23	16	77	2	2	False
4	5	31	17	40	2	4	False

In [129…

```
dff.columns
```

Out[129…

```
Index(['CustomerID', 'Age', 'Annual Income (k$)', 'Spending Score (1-100)',
       'Income Cluster', 'Spending and Income Cluster', 'Gender_Male'],
      dtype='object')
```

In [130…

```
dff = dff[['Age', 'Annual Income (k$)', 'Spending Score (1-100)', 'Gender_Male']]
dff.head()
```

Out[130…

	Age	Annual Income (k$)	Spending Score (1-100)	Gender_Male
0	19	15	39	True
1	21	15	81	True
2	20	16	6	False
3	23	16	77	False
4	31	17	40	False

In [131…

```
dff = scale.fit_transform(dff)
```

In [132…

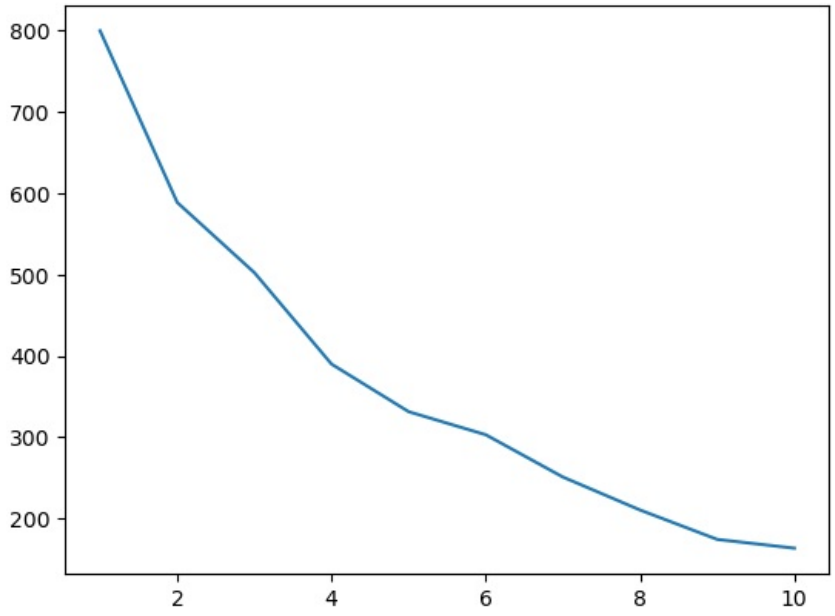
```
dff = pd.DataFrame(scale.fit_transform(dff))
```

In [133…

```
intertia_scores3=[]
for i in range(1,11):
    kmeans3=KMeans(n_clusters=i)
    kmeans3=KMeans(n_clusters=i)
    kmeans3.fit(dff)
    intertia_scores3.append(kmeans3.inertia_)
plt.plot(range(1,11),intertia_scores3)
#elbow at 4
```

Out[133…

```
[<matplotlib.lines.Line2D at 0x16635568f50>]
```



In [134…

```
df
```

Out[134…

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Income Cluster	Spending and Income Cluster
0	1	Male	19	15	39	2	4
1	2	Male	21	15	81	2	2
2	3	Female	20	16	6	2	4
3	4	Female	23	16	77	2	2
4	5	Female	31	17	40	2	4
...	...	...	...	...	...	...	...
195	196	Female	35	120	79	1	1
196	197	Female	45	126	28	1	3
197	198	Male	32	126	74	1	1
198	199	Male	32	137	18	1	3
199	200	Male	30	137	83	1	1

200 rows × 7 columns

In [135…

```
df.to_csv('Clustering.csv')
```